REVIEW

Bleeding management of thoracic trauma

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Purpose of review
Thoracic injuries are directly responsible for 20–30% of deaths in severe trauma patients and represent one of the main regions involved in preventable or potentially preventable deaths. Controlling bleeding in thoracic trauma is a major challenge because intrathoracic hemorrhagic lesions can lead to hemodynamic instability and respiratory failure.

Recent findings
The aim of managing intrathoracic hemorrhagic lesions is to control bleeding as quickly as possible and to control any respiratory distress. Extended focus assessment with sonography for trauma enables us to identify intrathoracic bleeding much more quickly and to determine the most appropriate therapeutic strategy.

Summary
Thoracic bleeding can result from the diaphragm, intrathoracic vessels (aorta, but also inferior or superior vena cava, and suprahepatic veins), lung, cardiac, or chest wall injuries. Depending on thoracic lesions (such as hemothorax or hemopericardium), hemodynamic instability, and respiratory failure, a pericardial window approach, sternotomy, thoracotomy, or emergency resuscitation thoracotomy may be considered after discussion with the surgeon. Alongside treatment of injuries, managing oxygenation, ventilation, hemodynamic, and coagulopathy are essential for the patient's outcome.

Keywords
bleeding management, extended focus assessment with sonography for trauma, managing oxygenation, thoracic trauma

INTRODUCTION

Thoracic injuries are directly responsible for 20–30% of deaths in severe trauma patients [1–3]. Several thoracic injuries can lead to life-threatening intrathoracic bleeding in severe trauma patients. Controlling intrathoracic bleeding must be part of a damage control strategy, which begins with prehospital care and continues in the hospital. The aim of the damage control strategy is to avoid preventable deaths (55% due to the inability to control bleeding). Among preventable (anatomic injuries clearly considered survivable if appropriate measures are taken) and potentially preventable (anatomic injuries that were severe but survivable under optimal care conditions) deaths (respectively, 5–11% and 25–43% of deaths), the thorax is the primary region injured (41%) [4,5]. This strategy involve a multidisciplinary team with emergency physicians, anesthetists, interventional radiologists, and surgeons. Most of these patients are managed nonoperatively with operative intervention being necessary in only 10–15% of patients with blunt thoracic trauma and 15–30% of penetrating thoracic trauma [6]. Intrathoracic hemorrhagic lesions can lead to hemodynamic instability and respiratory distress. The aim of managing intrathoracic hemorrhagic lesions is to control bleeding as quickly as possible and to address any respiratory failure.

ASSESSMENT OF INTRATHORACIC BLEEDING

Thoracic injuries are most often blunt injuries (mainly road traffic accidents and falls), but they can also be penetrating injuries (mainly by stab and gunshot wounds). Thoracic bleeding can result from diaphragm, intrathoracic vessels (aorta, but also inferior or superior vena cava, or suprahepatic veins), lungs, cardiac, and/or chest wall injuries

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KEY POINTS

- Thoracic injuries are directly responsible for 20–30% of deaths in severe trauma patients and represent one of the main regions involved in preventable or potentially preventable deaths.
- eFAST is crucial for rapidly determining the intrathoracic origin of hemodynamic instability.
- Echocardiography has 79% sensitivity and 92% specificity for detecting occult penetrating cardiac injuries in hemodynamically stable patients.
- Thoracic drainage with blood loss ≥1500 ml initially, or >200 ml/h over 2–4 h, should prompt discussion with a thoracic surgeon about performing a hemostasis thoracotomy.
- Depending on cardiac injury and hemodynamic instability, a pericardial window, thoracotomy, or emergency resuscitative thoracotomy may be discussed.

(Fig. 1). Identification of intrathoracic bleeding is based on clinical examination, extended focus assessment with sonography for trauma (eFAST), and whole-body computed tomography (CT) scan. This evaluation is an integral part of the classic assessment of severe trauma patients.

Clinical examination assesses airway freedom and detects respiratory and circulatory failures. In cases of thoracic trauma, the clinical examination assesses whether the thoracic trauma is open or closed, evaluates auscultatory asymmetry of breath sounds (after confirmation of correct tube position if the patient is mechanically ventilated), and looks for rib fractures with possible chest wall paradoxical movements (chest flail) and subcutaneous emphysema (indicative of pneumothorax). Distension of the jugular veins is a sign of compressive hemopneumothorax or pericardial tamponade (may be absent if hypovolemia is present). Beck's triad (hypotension, muffled heart sounds, and distension of the jugular vein) classically indicates pericardial tamponade. Beware, this triad, however, is not often present in traumatic tamponade [7]. Pulse asymmetry should be checked for aortic dissection. In cases of penetrating injury, entry wounds can be misleading (particularly with gunshot wounds), but a wound in the 'cardiac box' (bounded by the sternal notch at the top, the xiphoid process at the bottom, and the middle axillary lines laterally) should raise the suspicion of cardiac or large-vessel lesions.

eFAST is crucial to rapidly determine the intrathoracic origin of hemodynamic instability. eFAST is performed in search of hemothorax or pneumothorax (anterior views for pneumothorax and posterolateral views for hemothorax) or pericardial

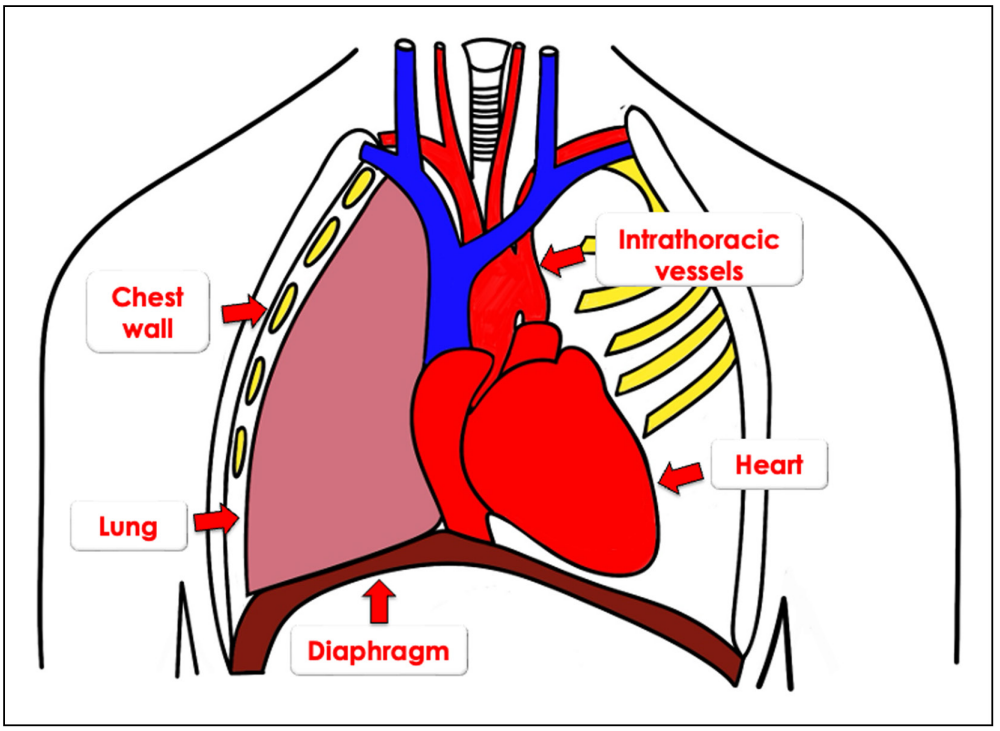


FIGURE 1. Potential sources of intrathoracic bleeding: diaphragmatic, intrathoracic vessels (aorta, but also inferior or superior vena cava, or suprahepatic veins), lung, cardiac, and/or chest wall injuries.

effusion (subxiphoid and/or anterior left intercostal views). Lack of lung sliding with observation of a lung point where lung sliding appears and disappears during respiration is highly specific for a pneumothorax [8] with diagnostic accuracy superior to chest radiography [9]. In addition, evaluation of lung sliding can confirm successful intubation and provide confirmation of lung isolation during one-lung ventilation. A hemothorax appears as an anechogenic (black) space between the diaphragm and the parietal pleura. eFAST is an effective method to diagnose pericardial fluid in trauma with 96% sensitivity, 98% specificity, and 97.5% accuracy for the presence of a pericardial effusion [10,11]. It helps diagnose pericardial tamponade by revealing evidence of diastolic collapse of the right ventricle, systolic collapse of the right atrium, and rightward deviation of the intraventricular septum [12]. It is worthwhile renewing eFAST as injuries may be revealed at a later stage, in particular, a pericardial effusion may initially be well tolerated and subsequently deteriorate with a risk of hemodynamic deterioration or even sudden cardiac arrest. The use of eFAST has thus changed the indications for radiology performed directly on the trauma bay [13]. Traditionally, chest X-ray was systematically performed on the trauma bay. With eFAST, however, chest X-ray could only be performed in cases of respiratory failure, if hemodynamic instability persists despite initial resuscitation, or if there is any uncertainty after eFAST examination.

Based on these assessments, the trauma leader may then choose:

(1) to perform chest drainage in the event of a compressive hemopneumothorax;
(2) to perform an orotracheal intubation to secure the airway and treat respiratory failure; or
(3) to insert an arterial catheter and a venous catheter in case of hemodynamic instability (in addition to two peripheral venous lines 14 G, 16 G). These catheters are inserted in the femoral position so that they can be installed simultaneously in a sterile fashion, thereby saving time. This site is also preferred, as it presents fewer risks than an approach in the superior vena cava territory, notably by avoiding the risks associated with cervical mobilization and the creation of a pneumothorax. The venous catheter (7–9 Fr) could thus be used to administer a vasopressor and to connect a high-flow blood and fluid warming system.

Then, in collaboration with the surgical teams and interventional radiologists, the trauma leader decides on the orientation of the patient according to the patient's failures, diagnostic suspected, and therapeutic priority (Fig. 2):

(1) Either a full-body CT scan is carried out to perform an exhaustive lesion assessment and identify any contrast leakages indicating active hemorrhage that can be treated by surgery or arteriography/embolization;
(2) Or it is decided to perform an emergency surgical hemostasis (pericardial window approach, thoracotomy, or emergency resuscitative thoracotomy) in the event of major hemodynamic instability unresponsive to fluid resuscitation and vasopressors.

MANAGEMENT OF HEMOTHORAX

Immediate pleural decompression is indicated in cases of life-threatening hemodynamic and/or respiratory instability associated with compressive hemothorax.

In the prehospital setting, needle decompression of a suffocating hemopneumothorax is a simple drainage procedure that can be effective (in 32–53% of cases). Needle thoracostomy, however, should be considered a last-resort technique (temporary relief measure) due to its high failure rate and the delay in achieving effective pleural decompression (risk of poor positioning, of puckering, of needle mobilization, and of insufficient drainage efficacy) [14]. Finger thoracostomy (without chest tube insertion) should be preferred to needle thoracentesis for urgent decompression [15,16].

Chest tube drainage should be considered once the patient is on the trauma bay. It is recommended to place large-caliber chest drains (28–36 Fr) in the mid-axillary line at the 4th or 5th costal space level (at the nipple) [13]. The chest tube should be placed in the trauma bay (i.e. before chest CT scan) in the case of hemodynamic and/or respiratory instability associated with a hemothorax. In the event of hemodynamic instability responding to initial resuscitation, and in the absence of respiratory distress, it is preferable to perform a chest CT scan before draining the thorax to exhaustively analyze thoracic injuries. Thoracic drainage with blood loss ≥1500 ml initially, or >200 mL/h over 2–4 h, should prompt discussion with a thoracic surgeon about performing a hemostasis thoracotomy [17].

MANAGEMENT OF TRANSFUSION

Blood product administration strategy is based on massive transfusion protocols (transfusion packs) and algorithms with target coagulation values

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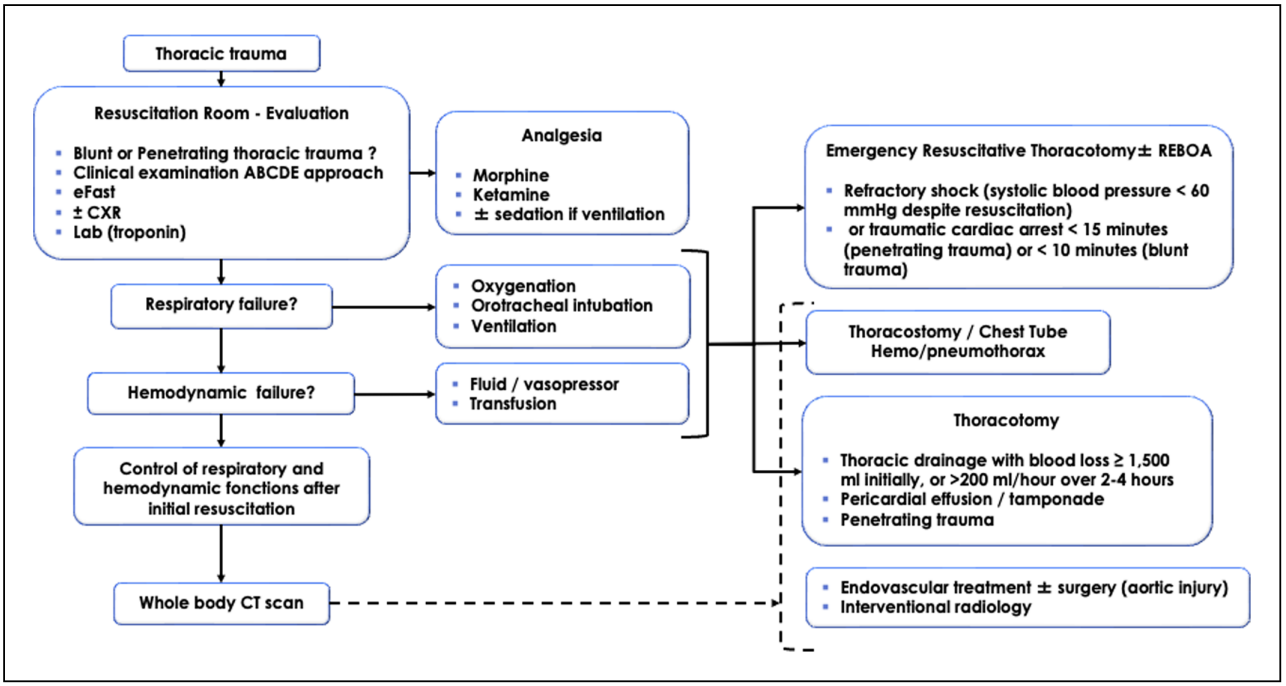


FIGURE 2. Hospital management of thoracic trauma. Depending on the severity of respiratory and/or hemodynamic failure, the trauma team may decide to perform either an emergency resuscitative thoracotomy, thoracic drainage, or thoracotomy. In the absence of respiratory or hemodynamic failure, or if failure has been controlled by initial resuscitation, a whole-body CT scan can be used to carry out an exhaustive assessment of thoracic lesions and detect active bleeding to discuss the most appropriate therapeutic orientation with the entire trauma team. CT, computed tomography.

derived from standard laboratory tests and/or bedside assessment of hemostasis, such as viscoelastic tests [18▪▪]:

(1) If hemorrhagic shock is strongly suspected based on prehospital assessment, it is advisable to order a 'pack' of group O red blood cells (thermostatically controlled container with temperature indicators so that it can be reused if not administered) before the trauma patient's arrival. These group O red blood cells can be combined with units of thawed plasma according to local procedures. Thus, on patient arrival, if hemodynamic instability is confirmed, transfusion is immediately started with group O red blood cells (Rh+ or Rh– if the woman is of childbearing age).
(2) Targeted hemoglobin is 7–9 g/dl with a fresh frozen plasma/red blood cells ratio ratio ≥ ½.
(3) Tranexamic acid should be administrated within the first 3 h of trauma (1 g over 10 min, then 1 g over 8 h).
(4) Administration of lyophilized plasma while awaiting the availability of thawed plasma can be an interesting strategy, particularly in situations where prehospital transport time is prolonged and the environment is hostile, such as in military contexts.
(5) Early platelet transfusion should be considered in the case of uncontrolled hemorrhage, to maintain a platelet count >50 g/l. In trauma brain injury patients, the platelet count should be raised to >100 g/l.
(6) Fibrinogen concentrates should be administrated in case of fibrinogenemia ≤1.5 g/l or viscoelastic tests in favor of fibrinogen deficiency (initial dose of 3–4 g is suggested).
(7) Ionized calcium levels should be monitored in the event of a massive transfusion and kept normal (targeted level: 1.1–1.2 mmol/l).
(8) Hypothermia should be controlled by rewarming of saline solutions and blood-derived products (via accelerator heater).

CARDIAC AND AORTIC INJURIES

Cardiac lesions are rare, but the mortality rate is high, with 50–90% of patients dying before reaching the hospital [19]. Cardiac injuries may present as hemorrhagic shock, hemothorax, and cardiac tamponade, but may also be accompanied by echocardiographic abnormalities, rhythm disorders (atrial fibrillation, supraventricular tachycardia, or ventricular tachycardia), EKG abnormalities (tachycardia, ST changes, and atrial fibrillation),

and/or troponin elevation. In penetrating trauma, a lesion in the cardiac box should raise suspicion of cardiac injury. Every thoracic or abdominal penetrating injury, however, should always raise the suspicion of cardiac injury, whatever the thoracic entry point(s) (i.e. even outside the cardiac box) [20]. Echocardiography is an essential diagnostic tool with 79% sensitivity and 92% specificity for detecting occult penetrating cardiac injuries in hemodynamically stable patients compared with the surgical technique of the pericardial window [10]. The accuracy of ultrasound, however, can be confounded by a left hemothorax, as this may signal a cardiac injury that is evacuating into the left thorax and thus may not have a pericardial effusion on ultrasound imaging. Echocardiography can also detect cardiac lesions such as valve anomalies, ventricular or atrial wall rupture, or chordae tendinae rupture. Depending on cardiac injury and hemodynamic instability, a pericardial window, a thoracotomy, or an emergency resuscitative thoracotomy may be discussed. Face to major heart failure, extracorporeal membrane oxygenation can be considered to stabilize patients to ensure tissue oxygenation and 'buy time' while surgical hemostasis is performed [21].

Thoracic vascular injuries are extremely challenging to treat, especially in the presence of hemorrhagic shock or circulatory failure. Their treatment requires a high degree of surgical expertise. For example, in the case of aortic trauma, which is most often due to blunt trauma (motor vehicle accidents or falls), initial mortality in the field is very high (estimated at 80–90% of all blunt traumatic aortic injuries) [22]. For patients arriving at the hospital, mortality is around 20–40%, mainly due to hemorrhagic shock (69%) [23,24]. This mortality is increased according to aortic injury severity, from 6% for grade 1 (intimal tear or intramural hematoma <10 mm) to 65% for grade 4 (aortic rupture). The timing of aortic rupture management depends on the patient's hemodynamic status at the time of admission. Unstable (due to this unstable aortic lesion) and in extremis patients should be taken to the operating room immediately [25]. For hemodynamically stable patients, if technically and anatomically feasible, endovascular stent grafting is preferred over the purely surgical approach in the management of ruptured thoracic aorta. Indeed, endovascular treatment is less invasive than the surgical approach, with fewer postoperative neurological deficits (paraplegia, stroke) and lower mortality [25,26]. In a recent multicenter retrospective study from the Traumabase, the French National Trauma Registry, the prevalence of blunt traumatic aortic injury on hospital admission was 1%, and of the patients treated, 92% benefited from endovascular treatment, 10% from open surgery, and 85% were treated within the first 24 h [24▪]. If surgery or endovascular treatment is not performed urgently, pharmacological control of blood pressure and heart rate is recommended with systolic arterial pressure < 120 mmHg and heart rate ≤60 bpm [27]. In the case of concomitant brain injury, maintaining a higher MAP is recommended. Intravenous beta-blockers (e.g. labetalol or esmolol) are recommended as first-line agents. If necessary, intravenous vasodilators (e.g. dihydropyridine calcium channel blockers or nitrates) may be added. Invasive monitoring with an arterial line is recommended [27].

INDICATIONS FOR EMERGENCY RESUSCITATIVE THORACOTOMY AND RESUSCITATIVE ENDOVASCULAR BALLOON OCCLUSION OF THE AORTA

Emergency resuscitation thoracotomy (ERT) has been described as a potentially lifesaving procedure for trauma patients with refractory shock or with recent loss of sign of life. The established role of ERT is the release of cardiac tamponade and the repair of cardiac injuries following thoracic trauma. Its place remains debated. The indication of ERT can be considered when refractory shock (systolic blood pressure < 60 mmHg despite resuscitation) or traumatic cardiac arrest <15 min (for penetrating trauma) or <10 min (for blunt trauma) with a negative abdominal focus assessment with sonography for trauma [28–31]. It is essential to follow a rational approach to select those patients who will benefit most from an ERT [29,30]. In prehospital care, it is not recommended [29,30]. In a recent French study based on the Traumabase [31▪], the rescue rate after ERT at 24 h and 28 days was 27 and 15%, respectively. In cases of refractory shock after penetrating trauma, survival was 64% at 24 h and 47% at 28 days.

Performing an ERT requires surgical expertise. The procedure always begins with a left anterolateral approach and may be extended by a transverse bi-thoracotomy (clamshell thoracotomy) or, more rarely and depending on the presumed origin of the hemorrhage, by a median sternotomy [32]. Ideally, this procedure requires a specialized cardiothoracic surgeon. Traumatic cardiac arrest, however, requires immediate management by the available trauma team, which may include surgeons with limited cardiothoracic experience. It is therefore essential to train surgeons involved in the management of trauma patients to perform an ERT.

Endovascular balloon occlusion of the aorta [resuscitative endovascular balloon occlusion of

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the aorta (REBOA)] using a balloon catheter via the femoral artery could be an attractive option for performing aortic clamping in the face of uncontrolled abdominal and pelvic bleeding. The trauma leader can decide on the level of occlusion (zone I, descending thoracic aorta; or zone III, above the aortic bifurcation) according to his or her assessment of the likely source of bleeding. REBOA can be used as a bridge to transfer the patient to the operating room for definitive hemostasis. Currently, REBOA is recommended for the management of subdiaphragmatic hemorrhage in patients who are unresponsive or only temporarily responsive to resuscitation, and for patients arriving at the hospital in cardiac arrest from trauma due to presumed life-threatening hemorrhage below the diaphragm [33]. Classically, REBOA is contraindicated in severe chest trauma [34], as aortic occlusion could increase thoracic arterial pressure and exacerbate any cardiac tamponade or intrathoracic hemorrhage. Two studies, however, have shown that REBOA zone 1 may be an effective alternative to ERT because it provides better or similar survival to ERT for patients requiring aortic occlusion in the setting of trauma, even in patients with severe chest trauma [35▪,36]. Recent publications, however, indicate that REBOA may be associated with increased mortality, delayed access to the operating room, and adverse effects such as deep vein thrombosis and limb ischemia [37–39▪]. The expertise of the team in charge of management is surely a determining factor in the respective efficacy of each technique and more data are needed to determine its exact role in thoracic injury management.

CONCLUSION

Thoracic injuries have the highest prevalence of serious injury in adult patients with major trauma. Deaths are linked to respiratory failure, hemorrhagic shock, and coagulopathy. Early identification of the origin of bleeding and multidisciplinary management of thoracic trauma are essential for good outcomes.

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Conflicts of interest

In recent years, J.D. has given lectures at symposia sponsored by LFB Health Care, Octapharma Company, and Edwards Lifesciences. For the remaining authors, there are no conflicts of interest.

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